
Formal Verification of the OpenVM RISC-V zkVM

Technical Report



NETHERMIND

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1 Executive Summary

This report provides an overview of the engagement performed by [Nethermind](#) on formally verifying, in the Lean proof assistant, a part of the [OpenVM RISC-V zkVM](#) against the [official RISC-V specification](#).

The work done in this engagement was based on the the official [OpenVM](#) repository (commit [645460f](#), dating from December 8th, 2025), and the engagement has resulted in the following three deliverables:

1. a mechanism for extracting Plonky3 constraints from OpenVM chips, described throughout §3 and publicly available at:
 - <https://github.com/NethermindEth/openvm> (OpenVM fork at commit [645460f](#)); and
 - <https://github.com/NethermindEth/openvm-stark-backend> (stark-backend fork at commit [972f5db](#));
2. a comprehensive verification infrastructure written in the [Lean proof assistant](#) (version 4.23.0-rc2), which is publicly available at github.com/NethermindEth/openvm-fv; and
3. as part of this infrastructure, Lean proofs that the OpenVM zk implementations of the following RISC-V opcodes satisfy their Lean RISC-V specifications:
 - 27 opcodes related to the functionalities of the ALU: ADD, ADDI, SUB, XOR, XORI, OR, ORI, AND, ANDI, SLT, SLTI, SLTU, SLTUI, SLL, SLLI, SRL, SRLI, SRA, SRAI, MUL, MULH, MULHU, MULHSU, DIV, DIVU, REM, and REMU;
 - 7 opcodes related to handling of control flow: AUIPC, BEQ, BNE, BLT, BGE, BLTU and BGEU; and
 - 2 opcodes related to memory manipulation: LOADW and STOREW.

During this process, we have not found bugs in the OpenVM implementation of the above-mentioned RISC-V opcodes. The verification was performed with respect to several assumptions and caveats, some of which are described in the document and all of which are summarised in §4.

Running the proofs. Assuming that an installation of Lean's lake package manager is present on the system, the infrastructure and all of the proofs can be built using:

```
lake update
lake exe cache get!
lake build
```

in the root folder of the repository, where the first two lines are superfluous but allow for a faster initial build.

2 Verification methodology

We discuss the key aspects of our methodology: formalisation of transpilation, the treatment of buses (both general and OpenVM-specific), and the workflow of per-opcode verification.

2.1 Transpilation

OpenVM [transpiles](#) RISC-V instructions into native OpenVM instructions, taking care of RISC-V specific requirements, such as the special treatment of register 0, in the process. This transpilation is formalised, for the opcodes covered by this engagement, in the [Transpiler.lean](#) file. As part of the transpilation, we formalise the conditions under which transpilation is successful. Specifically:

- as per [this](#) and [this](#) transpiler Rust code, that program counters are always divisible by 4; and
- as per [this](#) transpiler Rust code, that program counters are smaller than $\text{MAX_ALLOWED_PC} = 2^{30}$.

The formalisation of the transpiler is presented in more detail in the remainder of this report.

2.2 General treatment of buses

The OpenVM framework uses a number of buses for inter-chip communication and LogUp-based lookup-table information retrieval. A general bus entry over a field F is, in principle, of the form:

```
<multiplicity : F, data : Vector F>
```

consisting of a multiplicity (most commonly equalling 0, 1, or -1) and a vector of data, which can have different lengths and carry different meaning for different buses.

We formalise the notion of a bus as a Lean class, as follows:

```
/-- Type class for generic bus entries -/
class BusEntry (α : Type*) where
  multiplicity : α → F

  data_length : ℕ
  data : α → Vector F data_length

  axioms : α → Prop          -- Axioms on bus entries

  wf_properties : α → Prop -- Well-formedness properties of bus entries
  wf_assume_cond : α → Prop -- Condition under which well-formedness properties are assumed
  wf_assert_cond : α → Prop -- Condition under which well-formedness properties need to be asserted

  -- Assuming and proving of well-formedness properties
  assume (a : α) := wf_assume_cond a → wf_properties a
  assert (a : α) := wf_assert_cond a → wf_properties a

  -- Serialisation, deserialisation, properties
  serialise (x : α) := (multiplicity x, data x)
  serialiseToList (x : α) := (multiplicity x, (data x).toList)
  deserialise : F × Vector F data_length → α

  inv_ser_deser (x : F × Vector F data_length) : serialise (deserialise x) = x
  inv_deser_ser {x : α} : deserialise (serialise x) = x
```

providing the ability to specify:

- the multiplicity, the length of the data, and the data itself;
- the bus entry axioms, that is, the constraints that are enforced by the OpenVM implementation and/or the constraints that can only be inferred from multi-chip reasoning;
- the bus entry well-formedness (WF) properties, that is, the constraints that are assumed to hold for every bus entry a chip reads and must be proven to hold for every bus entry a chip writes;

- the conditions under which WF-properties should be assumed and asserted;
- how to generically serialise a bus entry into a tuple and a list; and
- how to deserialise a bus entry correctly.

Importantly, this approach is not OpenVM-specific and can be repurposed for at least all Plonky3-based zkVMs.

2.2.1 On bus balancing

For the results related to balancing of buses, we instantiate the field F to the BabyBear field¹ and equip the bus data (lists of BabyBear values) with a measure μ :

```
 $\mu : \text{List FBB} \rightarrow \mathbb{N}$ 
```

and define the notion of a *rising pair* with respect to μ as follows:

```
def rising_pair
  (dr ds : List FBB)
  (_ :  $\mu \text{ dr} < \mu \text{ ds}$ )
: List (FBB  $\times$  List FBB) :=
  [
    (-1, dr),
    ( 1, ds)
  ]
```

that is, as a pair for which the measure of the data that is put on the bus (ds) is strictly larger than the measure of the data that is read from the bus (dr). We define data read from the bus as the data with multiplicity -1 and the data put on the bus as the data with multiplicity 1 , as that is the case for the two relevant buses in OpenVM (the Execution and Memory buses, as discussed shortly).

Next, we define the notion of a *rising bus* as the bus whose data consists of rising pairs, as follows:

```
def rising_bus
  (bus_data : List (List FBB  $\times$  List FBB))
  (lt_proofs :  $\forall \text{ entry} \in \text{bus\_data}, \mu \text{ entry.1} < \mu \text{ entry.2}$ )
: List (FBB  $\times$  List FBB) :=
  List.flatten
    (List.map
      (fun (x : { y // y  $\in$  bus_data })  $\mapsto$ 
        let < dr, ds >, pf := x
        rising_pair  $\mu$  dr ds (lt_proofs (dr, ds) pf )) bus_data.attach)
```

2.2.2 The rising bus balancing theorem

The main bus balancing theorem that we prove is as follows:

```
lemma rising_bus_with_single_balancer_characterisation
  {ldata rdata : List FBB}
  (bus_data : List (List FBB  $\times$  List FBB))
  (lt_proofs :  $\forall \text{ entry} \in \text{bus\_data}, \mu \text{ entry.1} < \mu \text{ entry.2}$ )
  (h_not_empty :  $0 < \text{bus\_data.length}$ )
  (h_len_bus :  $([(1, \text{ldata})] ++ (\text{rising\_bus } \mu \text{ bus\_data lt\_proofs}) ++ [(-1, \text{rdata})]).\text{length} < \text{BB\_prime}$ )
  (h_balance : InteractionList.is_balanced
     $(([(1 : \text{FBB}), \text{ldata})] ++ (\text{rising\_bus } \mu \text{ bus\_data lt\_proofs}) ++ [((-1 : \text{FBB}), \text{rdata})])$ )
:
   $\exists \text{ xs}, \text{bus\_data.Perm } (\text{List.zip } (\text{ldata} :: \text{xs}) (\text{xs} ++ [\text{ldata}])) \wedge$ 
  List.Sorted (fun  $x_1 \ x_2 \mapsto \mu \ x_1 < \mu \ x_2$ )  $(\text{ldata} :: \text{xs} ++ [\text{ldata}])$ 
```

and it states that, if we have a non-empty rising bus with not more than $\text{BB_prime} - 2 = 2013265919$ pairs, and if this rising bus is balanced out by a pair of the form $[(1, \text{ldata}), (-1, \text{rdata})]$, then the measure of the rising bus is strictly increasing, the measure of ldata is the smallest measure of the bus, and the measure of rdata is the largest measure on the bus. We discuss how this theorem connects with the rest of the development in the following section.

¹The BabyBear field is the finite field used by OpenVM. Its size equals $\text{BB_prime} = 2013265921$.

2.3 The OpenVM buses

OpenVM makes use of the following six buses in the implementation of the RISC-V opcodes relevant for this engagement. These are identified by their index (a natural number), and are formalised in Lean as follows:

```
abbrev ExecutionBus : ℕ := 0
abbrev MemoryBus : ℕ := 1
abbrev RangeCheckerBus : ℕ := 4
abbrev ProgramBus : ℕ := 8
abbrev BitwiseBus : ℕ := 9
abbrev RangeTupleCheckerBus : ℕ := 11
```

We discuss each of them in the upcoming subsections. The important points to remember are that:

- the semantics of the bus entries is effectively given by the axioms and well-formedness properties; and
- by construction, the chips read contents from all six buses but write only to the Execution and Memory buses, which means that these are the only two buses for which we will need to prove well-formedness properties.

2.3.1 The Execution bus

```
structure ExecutionBusEntry where
  multiplicity : F
  pc : F
  timestamp : F

instance ExecutionBusEntryInstance : BusEntry FBB (ExecutionBusEntry FBB) := {
  ...
  axioms := fun ⟨multiplicity, pc, timestamp⟩ => ¬ multiplicity = 0 → pc < 2^30 ∧ pc % 4 = 0
  wf_assume_cond := fun ⟨mul, _, _⟩ => mul = -1,
  wf_assert_cond := fun ⟨mul, _, _⟩ => mul = 1,
  ...
}
```

which tells us that:

- the Execution bus entries carry two pieces of information: the program counter (pc) and the timestamp;
- the program counters are axiomatically declared to be smaller than 2^{30} and aligned to multiples of four; and
- there are no well-formedness properties for the entries of this bus.

Justification of axiomatic properties. The axiomatic properties in question are the properties guaranteed to hold for any program counter on the Program bus, as per the transpilation process. They are justified on paper as follows:

- E1)** every row of every chip puts on the execution bus either two zero-multiplicity entries or a pair of entries of the form $[\langle -1, [pc, timestamp] \rangle, \langle 1, [pc', timestamp'] \rangle]$, where $timestamp' > timestamp$;
- E2)** every chip obtains pc from the Program bus, from which we get, given transpiler properties, that $pc < 2^{30}$ and that $pc \% 4 = 0$, justifying the axiomatisation for the execution bus entry with multiplicity -1 ;
- E3)** the pairs of execution bus entries from (1) form a rising bus with respect to the timestamp component;
- E4)** this Rust code provides two additional entries that balance out the execution bus: in particular, these are $[\langle -1, [end_pc, end_timestamp] \rangle, \langle 1, [start_pc, start_timestamp] \rangle]$;
- E5)** this Rust code of the connector chip enforces that the start timestamp of the execution bus equals 1;
- E6)** this Rust code of the connector chip enforces that the end timestamp is smaller than 2^{29} , which, given (E1) and (E5), means that there cannot be more than 2^{29} pairs on the execution bus;
- E7)** given the previous points and the rising bus balancing theorem, we obtain that all of the pcs on the execution bus with multiplicity 1, *except potentially the one with the highest timestamp*, must have a matching pc with multiplicity -1 , which all originate from the Program bus and, therefore, satisfy the desired properties;
- E8)** this Rust code of the connector chip enforces that the pc with the highest timestamp can be looked up on the Program bus, which establishes the desired property for this pc as well, concluding the reasoning.

2.3.2 The Memory bus

The entry structure and semantics of the Memory bus are formalised as follows:

```

structure MemoryBusEntry where
  multiplicity : F
  as : F        -- address space
  ptr : F        -- memory address for 32-bit read/write
  x0 : F        -- first byte read/written
  x1 : F        -- second byte read/written
  x2 : F        -- third byte read/written
  x3 : F        -- fourth byte read/written
  timestamp : F -- timestamp

instance MemoryBusEntryInstance : BusEntry FBB (MemoryBusEntry FBB) := {
  ...
  axioms := fun ⟨multiplicity, _, _, _, _, _, timestamp⟩ => ¬ multiplicity = 0 → timestamp < 2 ^ 29
  wf_properties :=
    fun ⟨_, as, ptr, x0, x1, x2, x3, _⟩ =>
      as.val < 3 ^ ptr.val < 2 ^ 29 ^ x0.val < 256 ^ x1.val < 256 ^ x2.val < 256 ^ x3.val < 256
  wf_assume_cond := fun entry => entry.1 = -1,
  wf_assert_cond := fun entry => entry.1 = 1,
  ...
}

```

which tells us:

- that the RangeChecker bus entries carry seven pieces of information, described in code comments above;
- what the bounds on all of these seven parameters should be (for the timestamp axiomatically and for the others through the WF-properties); and
- that the WF-properties should be assumed/asserted if the multiplicity equals -1/1 (observed empirically).

The axiomatic property in question is justified on paper as follows:

- M1)** every row of every chip puts on the execution bus either two zero-multiplicity entries or a pair of entries of the form $\langle -1, [pc, timestamp] \rangle, \langle 1, [pc', timestamp'] \rangle$, where $timestamp' > timestamp$;
- M2)** if a given row of a given chip does not put non-zero multiplicity entries on the execution bus, then it also does not put non-zero multiplicity entries on the memory bus;
- M3)** every row of every chip puts on the memory bus either zero-multiplicity entries or pairs of entries of the form $\langle -1, [as, ptr, ..., timestamp_m] \rangle, \langle 1, [as, ptr, ..., timestamp'_m] \rangle$, where $timestamp'_m > timestamp_m$;
- M4)** for all $timestamp'_m$ from (M3), it holds that $timestamp'_m \in [timestamp, timestamp']$;
- M5)** by the balancing of the execution bus (M1)-(E4)-(E5)-(E6) and the rising bus balancing theorem, we obtain that the highest timestamp is the end timestamp of the trace;
- M6)** given (M5) and (E6), it must be that all timestamps are smaller than 2^{29} , which is the desired property.

On memory consistency. The pen-and-paper argument for memory consistency is as follows:

- C1)** for a given row of a given chip, and a fixed address space as and memory address ptr , the corresponding pairs of memory bus entries from (M3) form a rising bus with respect to the timestamp component;
- C2)** given (C1) and the bounds on the start and end timestamps (E5)-(E6), there can be no more than 2^{29} pairs on the memory bus overall;
- C3)** given (C1), (C2), and the rising bus balancing theorem, we obtain that the measures (that is, the timestamps) on the (rising) memory bus from (C1) are unique and strictly increasing;
- C4)** given (C3), we obtain that, on the memory bus from (C1), all reads (entries with multiplicity -1) read the last written value (or the initial value for the entry with the start timestamp), and that all writes (entries with multiplicity 1) come with a timestamp higher than the timestamp of the last read, which effectively means that memory accesses are consistent per address-space-memory-address pairs.

2.3.3 The RangeChecker bus

The entry structure and semantics of the RangeChecker bus are formalised as follows:

```

structure RangeCheckerBusEntry where
  multiplicity : F
  val : F
  deg : F

instance RangeCheckerBusEntryInstance : BusEntry FBB (RangeCheckerBusEntry FBB) := {
  ...
  axioms      := fun _      => True
  wf_properties := fun ⟨_, val, deg⟩ => deg.val < 31 ^ val.val < 2 ^ deg.val
  wf_assume_cond := fun ⟨mul, _, _⟩ => mul = 1,
  ...
}

```

which tells us that:

- the RangeChecker bus entries carry two pieces of information: deg and val;
- it holds that $\text{val} < 2^{\text{deg}}$ (hence the range-checking name of the bus); and
- the WF-properties should be assumed if multiplicity equals 1 (observed empirically).

2.3.4 The Program bus

The entry structure and semantics of the Program bus are formalised as follows:

```

def ProgramBusEntry.operand_properties (entry : ProgramBusEntry FBB) : Prop :=
  ∃ instruction data,
    (Transpiler.transpile_op instruction entry.multiplicity entry.pc = .some data) ^
    (ProgramBusEntry.deserialise FBB data = entry)

structure ProgramBusEntry where
  multiplicity : F
  pc : F
  opcode : F
  xa : F
  xb : F
  xc : F
  xd : F
  xe : F
  xf : F
  xg : F

instance ProgramBusEntryInstance
: BusEntry FBB (ProgramBusEntry FBB) :=
{
  ...
  axioms := fun _ => True
  wf_properties := ProgramBusEntry.operand_properties
  wf_assume_cond := fun entry => entry.1 = 1,
  wf_assert_cond := fun entry => entry.1 = -1,
  ...
}

```

which tells us that:

- the Program bus entries carry nine pieces of information: the program counter, pc; the opcode, opcode; and other parameters which denote, e.g., the registers from which data should be retrieved, immediate values, etc.;
- that the well-formedness properties are obtained from the transpilation of the opcode in question; and
- the WF-properties should be assumed if multiplicity equals 1 (observed empirically).

2.3.5 The Bitwise bus

The entry structure and semantics of the Bitwise bus are formalised as follows:

```

structure BitwiseBusEntry where
  multiplicity : F
  a : F
  b : F
  c : F
  op : F

instance BitwiseBusEntryInstance
: BusEntry FBB (BitwiseBusEntry FBB) :=
{
  ...
  axioms      := fun _           => True
  wf_properties := fun ⟨_, a, b, c, op⟩ => deg.val < 31 ∧ val.val < 2 ^ deg.val
  wf_assume_cond := fun ⟨mul, _, _⟩   => mul = 1,
  ...
}

```

which tells us that:

- the Bitwise bus entries carry four pieces of information: a, b, c, and op;
- it holds that:
 - $a.val < 256$, $b.val < 256$, and op equals either 0 or 1;
 - if $op = 0$, then $c.val = 0$; and
 - if $op = 1$, then $c.val = a.val \oplus b.val$;
- the WF-properties should be assumed if multiplicity equals 1 (observed empirically).

What is interesting about this bus is that the flexibility of the op parameter allows it to sometimes be used only as a range checker, and other times to also compute the exclusive disjunction.

2.3.6 The RangeTupleChecker bus

The entry structure and semantics of the RangeChecker bus are formalised as follows:

```

structure RangeTupleCheckerBusEntry where
  multiplicity : F
  x1 : F
  x2 : F

instance RangeTupleCheckerBusEntryInstance
: BusEntry FBB (RangeTupleCheckerBusEntry FBB) :=
{
  ...
  axioms      := fun _           => True
  wf_properties := fun ⟨_, x1, x2⟩ => x1 < 256 ∧ x2 < 2048
  wf_assume_cond := fun ⟨mul, _, _⟩ => mul = 1,
  ...
}

```

which tells us that:

- the RangeTupleChecker bus entries carry two pieces of information: x1 and x2;
- it holds that $x1 < 2^8 = 256$ and $x2 < 2^{11} = 2048$, range-checking both pieces of data at the same time; and
- the WF-properties should be assumed if multiplicity equals 1 (observed empirically).

We note that the range checks can, in principle, be instantiated by the user. It happens that, for the opcodes verified in this engagement, the upper bounds are set to 2^8 and 2^{11} .

2.4 Opcode verification workflow

For every verified opcode verified, we perform the following workflow in Lean, noting that there are parts of the workflow that are re-used for multiple opcodes, as OpenVM chips almost never implement a single opcode:

1. implement the transpilation of the opcode in our formalisation of the OpenVM transpiler;
2. formalise the core air of the chip that implements the opcode;
3. formalise the associated core air adapter, which connects the chip to the various buses;
4. formalise the air wrapper, which brings together the core air and its adapter;
5. extract the constraints and bus contents of the chip in question;
6. transform the extracted constraints and bus contents into human-readable form;
7. prove that the extracted constraints imply a pure human-readable RISC-V specification of the opcode, closely related to the behaviour specified in the official Lean RISC-V specification; and, finally
8. connect the structural part of the Lean RISC-V opcode specification to the pure specification of (5), effectively proving overall correctness of the opcode implementation.

3 An in-depth look into verification

In this section, we provide detailed, illustrative examples of the verification process. In particular: we show the verification workflow as described in §2.4 in detail for the ADD/ADDI opcodes (implemented by the base ALU chip), and also discuss the specificities of some other opcodes, such as branching opcodes, LOADW, and STOREW.

3.1 Example 1: ADD/ADDI

3.1.1 Step 1: Transpilation

As a first step, we extend the transpiler with support for the ADD and ADDI opcodes:

```
def transpile_op
  (inst : instruction) -- instruction, as per the Lean RISC-V spec
  (multiplicity : FBB) -- multiplicity
  (pc : FBB)           -- program counter
:
  Option (FBB × Vector FBB 9) -- Program bus entry
:=
  if pc % 4 = 0 ∧ pc < 2 ^ 30 then
    match inst with
    ...
    | .RTYPE (rs2, rs1, rd, rop.ADD) =>
      .some (if rd == regidx.Regidx 0
        then (multiplicity, #v[pc, 1, 0, 0, 0, 0, 0, 0, 0])
        else (multiplicity, #v[pc, 512, ind rd, ind rs1, ind rs2, 1, 1, 0, 0]))
    ...
    | .ITYPE (imm, rs1, rd, iop.ADDI) =>
      .some (if rd == regidx.Regidx 0
        then (multiplicity, #v[pc, 1, 0, 0, 0, 0, 0, 0, 0])
        else (multiplicity, #v[pc, 512, ind rd, ind rs1, utof (sign_extend_24 imm), 1, 0, 0, 0]))
```

where

```
def ind      (rd : regidx) : FBB      := ⟨4 * (regidx_to_fin rd).val, by omega⟩
def sign_extend_24 (bv : BitVec 12) : BitVec 24 := bv.signExtend 24
def utof      (bv : BitVec n) : FBB      := bv.toNat
```

noting in general:

- that the transpiler receives the instruction to transpile directly from the Lean RISC-V spec instruction type
- that the transpiler returns the corresponding bus entry that should be put on the Program bus; and
- how the transpilation branches on whether or not the target register is 0, in which case no operation is performed, mirroring the RISC-V spec;

and in particular:

- for ADDI, how the 12-bit immediate `imm` is sign-extended to 24-bits and then cast into the BabyBear field; the constraints of the base ALU chip further sign-extend this value to 32 bits.

3.1.2 Step 2: Core air

The ADD/ADDI opcodes are implemented as part of the [base ALU chip](#). In particular, the structural part of the base ALU core air is implemented in the Rust codebase as follows:

```
pub struct BaseAluCoreCols<T, const NUM_LIMBS: usize, const LIMB_BITS: usize> {
  pub a: [T; NUM_LIMBS],
  pub b: [T; NUM_LIMBS],
  pub c: [T; NUM_LIMBS],

  pub opcode_add_flag: T,
  pub opcode_sub_flag: T,
  pub opcode_xor_flag: T,
```

```
pub opcode_or_flag : T,
pub opcode_and_flag : T,
}
```

We transfer this structure to Lean, maintaining strong eyeball correspondence between the two definitions. We achieve this by implementing a series of macros as part of our [Lean zk-circuit infrastructure](#) that allow users to define air-related structures in a modular way. In particular, the two macros that are used for the generation of specialised circuit types are called `#define_air` and `#define_subair`: `#define_subair` is used for creating nested structure within a trace, whereas `#define_air` is used for generating the overall structure of the circuit.

In particular, the structure above is formalised in Lean, in the `BaseAluCoreAir.lean` file, as:

```
#define_subair "BaseAluCoreAir" using "openvm_encapsulation" where
  Column["a_0"]
  Column["a_1"]
  Column["a_2"]
  Column["a_3"]
  Column["b_0"]
  Column["b_1"]
  Column["b_2"]
  Column["b_3"]
  Column["c_0"]
  Column["c_1"]
  Column["c_2"]
  Column["c_3"]
  Column["opcode_add_flag"]
  Column["opcode_sub_flag"]
  Column["opcode_xor_flag"]
  Column["opcode_or_flag"]
  Column["opcode_and_flag"]
```

The correspondence between the two is almost one-to-one, with the only difference being that the Rust arrays are explicitly deconstructed per-index: that is, for example, the Rust `pub a: [T; NUM_LIMBS]`, as `NUM_LIMBS = 4` for 32-bit OpenVM, corresponds to the four Lean columns `a_0`, `a_1`, `a_2`, and `a_3`. From this macro, we then automatically generate two structures, `Raw_BaseAluCoreAir` and `Valid_BaseAluCoreAir`, as follows:

```
structure Raw_BaseAluCoreAir (F : Type) : Type
number of parameters: 1
fields:
  Raw_BaseAluCoreAir.columns : Nat -> Nat -> Nat -> F
  Raw_BaseAluCoreAir.a_0 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.a_1 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.a_2 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.a_3 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.b_0 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.b_1 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.b_2 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.b_3 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.c_0 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.c_1 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.c_2 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.c_3 : Nat -> Nat -> F
  Raw_BaseAluCoreAir.opcode_add_flag : Nat -> Nat -> F
  Raw_BaseAluCoreAir.opcode_sub_flag : Nat -> Nat -> F
  Raw_BaseAluCoreAir.opcode_xor_flag : Nat -> Nat -> F
  Raw_BaseAluCoreAir.opcode_or_flag : Nat -> Nat -> F
  Raw_BaseAluCoreAir.opcode_and_flag : Nat -> Nat -> F
constructor:
  Raw_BaseAluCoreAir.mk {F : Type} (columns : Nat -> Nat -> Nat -> F)
    (a_0 a_1 a_2 a_3 b_0 b_1 b_2 b_3 c_0 c_1 c_2 c_3 opcode_add_flag opcode_sub_flag
      opcode_xor_flag opcode_or_flag opcode_and_flag : Nat -> Nat -> F) :
    Raw_BaseAluCoreAir F
```

```
@[reducible] def Valid_BaseAluCoreAir : Type -> Type :=
```

```
fun F → { c // c.isValid }
```

where:

- the columns field purely represents a circuit (collection of traces): specifically, columns col row rotation is meant to denote the value in column col of row (row + rotation) % trace_length, which is enforced at the #define_air level, as illustrated shortly;
- the remaining 17 typed members of the structure represent the 17 columns, which are named as in the original Rust code: for example, a_0, opcode_add_flag, etc.; and
- the isValid predicate states the these 17 named columns (that is, sub-air members) are indeed equal to the appropriate locations in the appropriate traces, and, more generally, that any nested sub-airs are also valid.

Next, we look at the Rust code of the eval function of the base ALU chip and build a model of it in Lean. This amounts to defining and folding the definitions of various local variables used by eval. For example, in the Rust codebase there is the array carry_add that holds the carries from the addition:

```
let mut carry_add: [AB::Expr; NUM_LIMBS] = array::from_fn(|_| AB::Expr::ZERO);
let carry_divide = AB::F::from_canonical_usize(1 << LIMB_BITS).inverse();
...
for i in 0..NUM_LIMBS {
  ...
  carry_add[i] = AB::Expr::from(carry_divide) * (b[i] + c[i] - a[i] +
    if i > 0 { carry_add[i - 1].clone() } else { AB::Expr::ZERO });
  ...
}
```

a part of which is encoded into Lean as follows, again index-by-index:

```
def Valid_BaseAluCoreAir.carry_divide (c : Valid_BaseAluCoreAir FBB) : FBB := 2005401601

def Valid_BaseAluCoreAir.carry_add_0 (c : Valid_BaseAluCoreAir FBB) (row rotation : Nat) : FBB :=
  c.carry_divide * (c.b_0 row rotation + c.c_0 row rotation - c.a_0 row rotation)

def Valid_BaseAluCoreAir.carry_add_1 (c : Valid_BaseAluCoreAir FBB) (row rotation : Nat) : FBB :=
  c.carry_divide *
    (c.b_1 row rotation + c.c_1 row rotation - c.a_1 row rotation + c.carry_add_0 row rotation)
...

```

where the eyeball correspondence is once more straightforward, noting the above-mentioned row and rotation. This process is relatively simple to do by hand, but would be complex to do automatically as it would require reasoning about arbitrary rust code. The general strategy is to create a Lean definition for each let statement in the Rust constraints and a corresponding proof that the full expression is equal to this newly defined member. For the two carries shown above, these proofs are as follows:

```
@[openvm_encapsulation]
lemma BaseAluCoreAir.carry_add_0_def
  [Field F] (c : Valid_BaseAluCoreAir F) (row rotation: ℕ)
  :
    (2005401601 : F) * (c.b_0 row rotation + c.c_0 row rotation - c.a_0 row rotation) =
      c.carry_add_0 row rotation
:= rfl

@[openvm_encapsulation]
lemma BaseAluCoreAir.carry_add_1_def
  [Field F] (c : Valid_BaseAluCoreAir F) (row rotation: ℕ)
  :
    (2005401601 : F) * (c.b_1 row rotation + c.c_1 row rotation - c.a_1 row rotation +
      c.carry_add_0 row rotation) =
      c.carry_add_1 row rotation
:= rfl

```

Note how these proofs gets included in the set of those that the simplification proofs can use, denoted by `openvm_`-encapsulation, and now when the full expression appears in the constraint, it will get folded up into our newly defined member. The effect of this process is that the constraints will ultimately look almost exactly like the Rust assertions, making them easier to reason about.

3.1.3 Step 3: Core air adapter

The next step is to encode the adapter air for the base ALU chip. Its structural part is implemented in the Rust codebase as follows:

```
pub struct Rv32BaseAluAdapterCols<T> {
    pub from_state: ExecutionState<T>,
    pub rd_ptr: T,
    pub rs1_ptr: T,
    pub rs2: T,    /// Pointer if rs2 was a read, immediate value otherwise
    pub rs2_as: T, /// 1 if rs2 was a read, 0 if an immediate
    pub reads_aux: [MemoryReadAuxCols<T>; 2],
    pub writes_aux: MemoryWriteAuxCols<T, RV32_REGISTER_NUM_LIMBS>,
}
```

which, as we can see, contains three sub-airs, `from_state`, `from_state`, and `from_state`, together with several other columns. In Lean, this structure is encoded as follows, in the `Rv32BaseAluAdapterAir.lean` file:

```
#define_subair "Rv32BaseAluAdapterAir" using "openvm_encapsulation" where
  SubAir["from_state": "ExecutionState" width := 2]
  Column["rd_ptr"]
  Column["rs1_ptr"]
  Column["rs2"]
  Column["rs2_as"]
  SubAir["reads_aux_0": "MemoryReadAuxCols" width := 3]
  SubAir["reads_aux_1": "MemoryReadAuxCols" width := 3]
  SubAir["writes_aux": "MemoryWriteAuxCols_4" width := 7]
```

where the auxiliary airs, `ExecutionState`, `MemoryReadAuxCols`, and `MemoryWriteAuxCols_4` are defined separately in `ExecutionState.lean` and `Memory.lean`. This particular encoding above showcases how easy is it to combine airs in our verification infrastructure, mirroring their Rust implementations.

3.1.4 Step 4: Base ALU air wrapper

The last structural step is to connect the core air with the adapter air using the `VmAirWrapper` mechanism, implemented in the Rust codebase as follows:

```
pub type Rv32BaseAluAir =
  VmAirWrapper<Rv32BaseAluAdapterAir, BaseAluCoreAir<RV32_REGISTER_NUM_LIMBS, RV32_CELL_BITS>>;
```

and encoded in Lean as follows (in the `Alu/VmAirWrapper_alu.lean` file):

```
#define_air "VmAirWrapper_alu" using "openvm_encapsulation" where
  MainSubAir["adapter": "Rv32BaseAluAdapterAir" width := 19]
  MainSubAir["core": "BaseAluCoreAir" width := 17]
```

which results, analogously to the sub-air definitions above, in an automatic creation of the following Lean structure representing an air:

```
structure Raw_VmAirWrapper_alu (F ExtF : Type) : Type
number of parameters: 2
fields:
  Raw_VmAirWrapper_alu.buses : Nat -> List (F x List F)
  Raw_VmAirWrapper_alu.challenge : Nat -> ExtF
  Raw_VmAirWrapper_alu.exposed : Nat -> ExtF
  Raw_VmAirWrapper_alu.main : Nat -> Nat -> Nat -> F
  Raw_VmAirWrapper_alu.permutation : Nat -> Nat -> Nat -> ExtF
  Raw_VmAirWrapper_alu.preprocessed : Nat -> Nat -> Nat -> F
  Raw_VmAirWrapper_alu.public_values : Nat -> F
```

```

Raw_VmAirWrapper_alu.last_row : Nat
Raw_VmAirWrapper_alu.adapter : Raw_Rv32BaseAluAdapterAir F
Raw_VmAirWrapper_alu.core : Raw_BaseAluCoreAir F
constructor:
Raw_VmAirWrapper_alu.mk {F ExtF : Type}
  (buses : Nat -> List (F x List F)) (challenge_exposed : Nat -> ExtF)
  (main : Nat -> Nat -> Nat -> Nat -> F) (permutation : Nat -> Nat -> Nat -> ExtF)
  (preprocessed : Nat -> Nat -> Nat -> F) (public_values : Nat -> F)
  (last_row : Nat) (adapter : Raw_Rv32BaseAluAdapterAir F) (core : Raw_BaseAluCoreAir F) :
  Raw_VmAirWrapper_alu F ExtF

```

where the most important fields for this engagement, in addition to the adapter and the core, are:

- buses, which contains the data that is put on the buses by the base ALU chip;
- main, which contains the main trace of the base ALU chip;
- last_row, which tells us how many rows the main trace has, in the sense that the above-mentioned trace_length corresponds to last_row + 1.

as well as the wrapper Valid_VmAirWrapper_alu structure:

```

@[reducible]
def Valid_VmAirWrapper_alu : Type -> Type -> Type :=
fun F ExtF => { c // c.isValid }

```

where the is_Valid predicate states, in addition to sub-air validity and column matching, that the rotation parameter behaves as expected.

This air definition is accompanied by the manually written definitions and proofs that connect the columns of the base ALU core air to the columns of the adapter air as per the Rust code, such as, for example:

```

let rs2_imm =
  rs2_limbs[0].clone() +
  rs2_limbs[1].clone() * AB::Expr::from_canonical_usize(1 << RV32_CELL_BITS) +
  rs2_sign.clone() * AB::Expr::from_canonical_usize(1 << (2 * RV32_CELL_BITS));

```

which is encoded in Lean as follows:

```

def Valid_VmAirWrapper_alu.rs2_imm
  [Field ExtF]
  (c : Valid_VmAirWrapper_alu FBB ExtF)
  (row rotation : Nat)
: FBB :=
  c.rs2_limbs row rotation 0 + c.rs2_limbs row rotation 1 * 256 + c.rs2_sign row rotation * 65536

```

3.1.5 Step 5: Constraint extraction

The next file of relevance is [Extraction/VmAirWrapper_alu.lean](#), which is generated with the help of our constraint extraction system and contains all of the extracted main-trace constraints, permutation-related constraints, and the contents placed on all of the buses by the base ALU chip.

In particular, we extract the constraints by modifying the stark-backend crate that OpenVM uses. Precisely, in the `AirKeygenBuilder::generate_pk` method, the `SymbolicRapBuilder` is already being used to generate symbolic constraints for the air in order to calculate the degree of those constraints. This `SymbolicRapBuilder` also creates a list of symbolic bus interactions generated during the evaluation of the air. These symbolic constraints and bus interactions are in the form of Abstract Syntax Trees for arithmetic expressions. We hook into this process, taking the generated symbolic constraints and interactions and printing them to the standard output so that they can be retrieved without having to further modify the control flow of the crate. We insert this logic into the stark-backend crate so that it can easily be run by simply running the tests for each circuit.

From the arithmetic-expression ASTs obtained from the stark backend, we create a purposefully simple and verbose flattening of each AST into Lean arithmetic expressions, avoiding any optimisation or simplification which could introduce bugs, preferring instead to allow Lean to do such things verifiably. This is done for all generated constraints first, and then for the bus interactions. As the permutation trace (that is, the trace in the extension field) is only used

in these circuits for the constraints generated to enforce interactions, the extractor takes all generated constraint expressions and converts any that reference the permutation trace into comments. Again, great care is taken to make sure that any bugs would be immediately obvious, in the sense that constraints are commented out rather than deleted so that we can double check that only the correct constraints are being ignored.

When it comes to bus interactions, we create a list representation that includes all bus interactions specialised to coming from the first row, then the second, etc. This simplifies reasoning as we tend to reason about one row at a time. We also need to separate the interactions into the individual buses, which is done by constructing a hash map keyed on bus index and first inserting all interactions into there. As seen earlier, bus indices are left purely as integers, again for the sake of reducing processing done outside of Lean.

We illustrate the outcome of the extraction process using the constraint arising from the last (fourth) iteration of the following loop:

```
for i in 0..NUM_LIMBS {
  ...
  carry_sub[i] = AB::Expr::from(carry_divide)
  * (a[i] + c[i] - b[i] +
    if i > 0 { carry_sub[i - 1].clone() } else { AB::Expr::ZERO });
  builder.when(cols.opcode_sub_flag).assert_bool(carry_sub[i].clone());
}
```

which states that, when the opcode_sub_flag is non-zero, carry_sub[3] equals either zero or one. This constraint, which happens to be the 13th one, gets extracted to the following Lean code:

```
def constraint_13
{C : Type → Type → Type}
{F ExtF : Type} [Field F] [Field ExtF]
[Circuit F ExtF C]
(c : C F ExtF) (row : Nat)
:=
((Circuit.main c (id := 0) (column := 32) (row := row) (rotation := 0)) * ((2005401601 *
(((Circuit.main c (id := 0) (column := 22) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 30) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 26) (row := row) (rotation := 0))) + (2005401601 *
(((Circuit.main c (id := 0) (column := 21) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 29) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 25) (row := row) (rotation := 0))) + (2005401601 *
(((Circuit.main c (id := 0) (column := 20) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 28) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 24) (row := row) (rotation := 0))) + (2005401601 *
(((Circuit.main c (id := 0) (column := 19) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 27) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 23) (row := row) (rotation := 0))) + 0)))))) *
((2005401601 * (((Circuit.main c (id := 0) (column := 22) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 30) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 26) (row := row) (rotation := 0))) +
(2005401601 * (((Circuit.main c (id := 0) (column := 21) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 29) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 25) (row := row) (rotation := 0))) +
(2005401601 * (((Circuit.main c (id := 0) (column := 20) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 28) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 24) (row := row) (rotation := 0))) +
(2005401601 * (((Circuit.main c (id := 0) (column := 19) (row := row) (rotation := 0)) +
(Circuit.main c (id := 0) (column := 27) (row := row) (rotation := 0))) -
(Circuit.main c (id := 0) (column := 23) (row := row) (rotation := 0))) + 0)))))) - 1))) = 0
```

where, as can be seen, all of the variable definitions have been fully unfolded.

On the other hand, the contents of the buses is described as follows (for readability only showing the contents of the Program bus, whose index is 8, in full):

```
def constrain_interactions
{C : Type → Type → Type}
{F ExtF : Type} [Field F] [Field ExtF]
```

```

[Circuit F ExtF C]
(c : C F ExtF)
:=
Circuit.buses c = fun index =>
  -- Execution bus
  if index = 0 then (List.range (Circuit.last_row c + 1)).flatMap (fun row => [ ... ])
  -- Memory bus
  else if index = 1 then (List.range (Circuit.last_row c + 1)).flatMap (fun row => [ ... ])
  -- Range-checking bus
  else if index = 4 then (List.range (Circuit.last_row c + 1)).flatMap (fun row => [ ... ])
  -- Program bus
  else if index = 8 then (List.range (Circuit.last_row c + 1)).flatMap (fun row =>
    [
      (((((0 + (Circuit.main c (id := 0) (column := 31) (row := row) (rotation := 0))) +
        (Circuit.main c (id := 0) (column := 32) (row := row) (rotation := 0))) +
        (Circuit.main c (id := 0) (column := 33) (row := row) (rotation := 0))) +
        (Circuit.main c (id := 0) (column := 34) (row := row) (rotation := 0))) +
        (Circuit.main c (id := 0) (column := 35) (row := row) (rotation := 0))),
      [
        (Circuit.main c (id := 0) (column := 0) (row := row) (rotation := 0)),
        (512 + (((((0 + ((Circuit.main c (id := 0) (column := 31) (row := row) (rotation := 0)) * 0)) +
          ((Circuit.main c (id := 0) (column := 32) (row := row) (rotation := 0)) * 1)) +
          ((Circuit.main c (id := 0) (column := 33) (row := row) (rotation := 0)) * 2)) +
          ((Circuit.main c (id := 0) (column := 34) (row := row) (rotation := 0)) * 3)) +
          ((Circuit.main c (id := 0) (column := 35) (row := row) (rotation := 0)) * 4))),
        (Circuit.main c (id := 0) (column := 2) (row := row) (rotation := 0)),
        (Circuit.main c (id := 0) (column := 3) (row := row) (rotation := 0)),
        (Circuit.main c (id := 0) (column := 4) (row := row) (rotation := 0)),
        1,
        (Circuit.main c (id := 0) (column := 5) (row := row) (rotation := 0)),
        0,
        0
      ]
    ]
  )
  -- Bitwise bus
  else if index = 9 then (List.range (Circuit.last_row c + 1)).flatMap (fun row => [ ... ])
  -- Other buses
  else []

```

All of the constraints are pasted into a template file in the Lean project for this basic extraction output, yielding the `Extraction/VmAirWrapper_alu.lean` file. At this level, the `Circuit` structure referenced in the expressions (which contains the various traces) is completely generic and so the only human interaction is to name the namespace and simp categories for these constraints. Also, at this point, the logic of the constraints is entirely in Lean.

3.1.6 Step 6: Air wrapper with human-readable constraints

The next step is to recover human-readable information from the extracted constraints and bus contents, which we do in the `Constraints/VmAirWrapper_alu.lean` file. Firstly, with the help of our simplification infrastructure, we address all of the constraints. For example, we prove that the fully unfolded constraint 13 from the previous section (`VmAirWrapper_alu.extraction.constraint_13 air row`) is equivalent to its human-readable form (`constraint_13 air row`) as follows:

```

@[VmAirWrapper_alu_constraint_and_interaction_simplification]
def constraint_13 (air : Valid_VmAirWrapper_alu F ExtF) (row : Nat) : Prop :=
  air.core.opcode_sub_flag row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 1

@[VmAirWrapper_alu_air_simplification]
lemma constraint_13_of_extraction
  (air : Valid_VmAirWrapper_alu F ExtF) (row : Nat)
  :
  VmAirWrapper_alu.extraction.constraint_13 air row <=> constraint_13 air row
:= by
  simp_all [openvm_encapsulation,

```



```
VmAIRWrapper_alu_constraint_and_interaction_simplification]
```

Importantly, these simplification lemmas and their proofs are auto-generated with placeholders for the simplified constraints, and then these placeholders can be revealed by running the proofs. Using the example at hand, we would have:

```
@[VmAIRWrapper_alu_constraint_and_interaction_simplification]
def constraint_13 (air : Valid_VmAIRWrapper_alu F ExtF) (row : N) : Prop :=
  sorry
```

and the above lemma. Its proof will fail, but the state after the simplifications

```
F ExtF : Type
inst11 : Field F
inst† : Field ExtF
air : Valid_VmAIRWrapper_alu F ExtF
row : N
⊢ air.core.opcode_sub_flag row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 1 ↔ sorry
```

tells us that constraint_13 should equal

```
air.core.opcode_sub_flag row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 0 ∨ air.core.carry_sub_3 row 0 = 1
```

When it comes to the buses, we also restate their contents in more readable terms, as follows:

```
@[VmAIRWrapper_alu_constraint_and_interaction_simplification]
def constrain_interactions (air : Valid_VmAIRWrapper_alu FBB ExtF) : Prop :=
  air.buses = fun index |->
    if index = ExecutionBus then (List.range (air.last_row + 1)).flatMap (executionBus_row air)
    else if index = MemoryBus then (List.range (air.last_row + 1)).flatMap (memoryBus_row air)
    else if index = RangeCheckerBus then (List.range (air.last_row + 1)).flatMap (rangeCheckerBus_row air)
    else if index = Program then (List.range (air.last_row + 1)).flatMap (programBus_row air)
    else if index = BitwiseBus then (List.range (air.last_row + 1)).flatMap (bitwiseBus_row air)
    else []
```

where, for instance, the contents of the Program bus are given by:

```
def programBus_row [Field ExtF]
  (air : Valid_VmAIRWrapper_alu FBB ExtF)
  (row : Nat)
: List (FBB × List FBB) :=
[
  air.core.is_valid row 0,
  [
    air.adapter.from_state.pc row 0,
    (air.core.ctx row 0).instruction.opcode,
    air.adapter.rd_ptr row 0,
    air.adapter.rs1_ptr row 0,
    air.adapter.rs2 row 0,
    1,
    air.adapter.rs2_as row 0,
    0,
    0
  ]
]
```

which can be matched directly against the fully unfolded contents from the previous section. For example, given the following Rust code:

```
let flags = [
  cols.opcode_add_flag, cols.opcode_sub_flag, cols.opcode_xor_flag,
  cols.opcode_or_flag, cols.opcode_and_flag,
];

let is_valid = flags.iter().fold(AB::Expr::ZERO, |acc, &flag| {
```

```
...
acc + flag.into()
});
```

and the definition of the airs at hand, we have that

```
air.core.is_valid row 0
```

corresponds precisely to

```
(((((0 + (Circuit.main c (id := 0) (column := 31) (row := row) (rotation := 0))) +
  (Circuit.main c (id := 0) (column := 32) (row := row) (rotation := 0))) +
  (Circuit.main c (id := 0) (column := 33) (row := row) (rotation := 0))) +
  (Circuit.main c (id := 0) (column := 34) (row := row) (rotation := 0))) +
  (Circuit.main c (id := 0) (column := 35) (row := row) (rotation := 0)))
```

The `Constraints/VmAirWrapper_alu.lean` file further contains lemmas that establish equivalence of unfolded and human-readable constraints and bus contents, as well as some basic properties about the constraints, such as that at most one opcode can be non-zero and that the chip opcodes must be in a certain range.

Axioms, assumptions, assertions. Finally, we define, based on our approach to bus entries as described in 2.2, the related axioms, assumptions, and assertions per row of the base ALU chip, as follows:

```
-- Axiomatic properties
def axioms [Interaction.BusEntry FBB a] (rowData : List a) : Prop :=
  List.Forall id (rowData.map (Interaction.BusEntry.axioms FBB))

-- Well-formedness properties to be assumed
def propertiesToAssume [Interaction.BusEntry FBB a] (rowData : List a) : Prop :=
  List.Forall id (rowData.map (Interaction.BusEntry.assume FBB))

-- Well-formedness properties to be asserted
def propertiesToAssert [Interaction.BusEntry FBB a] (rowData : List a) : Prop :=
  List.Forall id (rowData.map (Interaction.BusEntry.assert FBB))

-- All of the axiomatic properties per row
def axiomsPerRow [Field ExtF]
  (air : Valid_VmAirWrapper_alu FBB ExtF) (row : Nat)
: Prop :=
  axioms (_executionBus_row air row) /\ axioms (_memoryBus_row air row) /\
  axioms (_rangeCheckerBus_row air row) /\ axioms (_programBus_row air row) /\
  axioms (_bitwiseBus_row air row)

-- All of the well-formedness properties to be assumed per row
def wf_propertiesToAssumePerRow [Field ExtF] (air : Valid_VmAirWrapper_alu FBB ExtF) (row : Nat)
: Prop :=
  propertiesToAssume (_executionBus_row air row) /\ propertiesToAssume (_memoryBus_row air row) /\
  propertiesToAssume (_rangeCheckerBus_row air row) /\
  propertiesToAssume (_programBus_row air row) /\ propertiesToAssume (_bitwiseBus_row air row)

-- All of the well-formedness properties to be asserted per row
def wf_propertiesToAssertPerRow [Field ExtF] (air : Valid_VmAirWrapper_alu FBB ExtF) (row : Nat)
: Prop :=
  propertiesToAssert (_executionBus_row air row) /\ propertiesToAssert (_memoryBus_row air row) /\
  propertiesToAssert (_rangeCheckerBus_row air row) /\
  propertiesToAssert (_programBus_row air row) /\ propertiesToAssert (_bitwiseBus_row air row)
```

3.1.7 Step 7: Pure RISC-V specification

As part of our next step, in the `Spec/BaseALU.lean` file, we prove that, for non-padding rows (that is, the rows for which it holds that `air.core.is_valid row 0 = 1`):

- if the values read from the buses satisfy the desired properties, then the values put on the buses also satisfy these properties; and
- the constraints imply the desired human-readable specification of the five base ALU opcodes, both for immediate and non-immediate variants.

For these proofs, we make use of the following hypotheses:

```
variable
  (ExtF : Type) [Field ExtF]
  (air : Valid_VmAirWrapper_alu FBB ExtF)
  (row : Nat)
  (row_in_range : row <= air.last_row)
  (constraints : VmAirWrapper_alu.constraints.allHold air row row_in_range)
  (row_valid : air.core.is_valid row 0 = 1)
  (assumptions : axiomsPerRow air row)
  (propertiesToAssume : wf_propertiesToAssumePerRow air row)
```

Bus content properties. First, we prove that the properties that need to be asserted about the content that is put on the buses by the base ALU chip actually hold:

```
include
  row_valid constraints axioms propertiesToAssume
in
  lemma wf_propertiesToAssert : wf_propertiesToAssertPerRow air row
```

Out of these properties, the only one that is noteworthy is the one that states that the four values denoting the result of executing the opcode are, in fact, bytes:

```
(air.core.a_0 row 0).val < 256 /\ (air.core.a_1 row 0).val < 256 /\
(air.core.a_2 row 0).val < 256 /\ (air.core.a_3 row 0).val < 256
```

Human-readable RISC-V specification. From this point on, we no longer treat the chip as a whole, focusing instead on the specific opcodes implemented by the chip, but still For this example, we focus on the ADD and ADDI opcodes, noting that the related correctness theorems are still phrased generally for all of the base ALU opcodes, given that a substantial amount of the reasoning is the same.

The theorem that covers the ADD opcode is as follows:

```
/-- From non-immediate ALU opcode to RISC-V opcode -/
def rop_of_ALU_opcode (opcode : FBB) : rop :=
  if opcode = 512 then .ADD else
  if opcode = 513 then .SUB else
  if opcode = 514 then .XOR else
  if opcode = 515 then .OR else .AND

include
  row_valid constraints axioms propertiesToAssume
in
  /-- The non-immediate variants of the five base ALU opcodes are implemented as per the RISC-V spec --/
  theorem spec_base_ALU_non_imm
    (_ : air.adapter.rs2_as row 0 = 1)
  :
    U32.toBV #v[(air.core.a_0 row 0).val,
      (air.core.a_1 row 0).val,
      (air.core.a_2 row 0).val,
      (air.core.a_3 row 0).val]
    =
    execute_RTYPE_pure
      (U32.toBV #v[(air.core.b_0 row 0).val,
        (air.core.b_1 row 0).val,
        (air.core.b_2 row 0).val,
        (air.core.b_3 row 0).val])
      (U32.toBV #v[(air.core.c_0 row 0).val,
```

```

      (air.core.c_1 row 0).val,
      (air.core.c_2 row 0).val,
      (air.core.c_3 row 0).val])
    (rop_of_ALU_opcode (air.core.ctx row 0).instruction.opcode)

```

where `execute_RTYPE_pure` captures the pure part of the executing RTYPE opcodes as per the Lean RISC-V spec:

```

/-- Pure part of 32-bit 'execute_RTYPE' -/
def execute_RTYPE_pure (op1 : BitVec 32) (op2 : BitVec 32) (op : rop) :=
  match op with
  | .ADD => op1 + op2
  | .SUB => op1 - op2
  | .XOR => op1 ^^^ op2
  | .OR  => op1 ||| op2
  | .AND => op1 &&& op2
  | ...

```

Within the proof of the theorem, we branch on the five opcodes of the base ALU, providing a separate proof for each. The ADD-specific goal is as follows:

```

⊢ U32.toBV
  #v[BitVec.ofNat 8 ↑(air.core.a_0 row 0), BitVec.ofNat 8 ↑(air.core.a_1 row 0),
    BitVec.ofNat 8 ↑(air.core.a_2 row 0), BitVec.ofNat 8 ↑(air.core.a_3 row 0)] =
  U32.toBV
  #v[BitVec.ofNat 8 ↑(air.core.b_0 row 0), BitVec.ofNat 8 ↑(air.core.b_1 row 0),
    BitVec.ofNat 8 ↑(air.core.b_2 row 0), BitVec.ofNat 8 ↑(air.core.b_3 row 0)] +
  U32.toBV
  #v[BitVec.ofNat 8 ↑(air.core.c_0 row 0), BitVec.ofNat 8 ↑(air.core.c_1 row 0),
    BitVec.ofNat 8 ↑(air.core.c_2 row 0), BitVec.ofNat 8 ↑(air.core.c_3 row 0)]

```

and is discharged by case analysis on the carries:

```

simp_all [← BitVec.toNat_inj, U32.toBV_toNat, U32.toNat]
clear * - add_0 add_1 add_2 add_3 ub_a0 ub_a1 ub_a2 ub_a3
      ub_b0 ub_b1 ub_b2 ub_b3 ub_c0 ub_c1 ub_c2 ub_c3
rcases add_0 <|> rcases add_1 <|>
rcases add_2 <|> rcases add_3 <|>
simp_all <|> omega

```

The proof of ADDI is done as part of an analogous theorem, targeting the appropriate immediate variants of the base ALU opcodes (ADDI, XORI, ORI, and ANDI):

```

/-- From immediate ALU opcode to RISC-V opcode -/
def iop_of_ALU_opcode (opcode : FBB) : iop :=
  if opcode = 512 then .ADDI else
  if opcode = 514 then .XORI else
  if opcode = 515 then .ORI else .ANDI

include
  row_valid constraints axioms propertiesToAssume
in
  /-- The immediate variants of the five base ALU opcodes are implemented as per the RISC-V spec -/
  theorem spec_base_ALU_imm
    (h_imm : air.adapter.rs2_as row 0 = 0)
  :
    U32.toBV #v[(air.core.a_0 row 0).val,
      (air.core.a_1 row 0).val,
      (air.core.a_2 row 0).val,
      (air.core.a_3 row 0).val]
    =
    execute_ITYPE_pure
      (BitVec.ofNat 12 (air.adapter.rs2 row 0).val)
      (U32.toBV #v[(air.core.b_0 row 0).val,
        (air.core.b_1 row 0).val,

```

```
(air.core.b_2 row 0).val,
  (air.core.b_3 row 0).val])
(iop_of_ALU_opcode (air.core.ctx row 0).instruction.opcode)
```

where `execute_ITYPE_pure` corresponds to the pure part of the execution of the ITYPE opcodes as per the official Lean RISC-V spec:

```
/-- Pure part of 32-bit 'execute_ITYPE' -/
def execute_ITYPE_pure (imm : BitVec 12) (op1 : BitVec 32) (op : iop) :=
  let immext := sign_extend (m := 32) imm
  execute_RTYPE_pure op1 immext (rop_of_iop op)
```

3.1.8 Step 8: Equivalence with RISC-V specification

The final step in proving correctness of opcode implementation is to connect the associated constraints with the structural part of the RISC-V specification.

Opcode input/output/behaviour. First, we isolate the input, the output, and the behaviour of each opcode. For ADD, this is done in the `add.lean` file, through which we go in detail, giving explanations in comments:

```
-- Input required for executing the ADD opcode
structure AddInput where
  r1_val : BitVec 32 -- first operand of ADD
  r2_val : BitVec 32 -- second operand of ADD
  rd      : Fin 32   -- intended output register (between 0 and 31 inclusive)
  PC      : BitVec 32 -- current program counter

-- Output of the ADD opcode
structure AddOutput where
  nextPC : BitVec 32 -- next program counter
  rd      : Option (Finset.Icc 1 31 × BitVec 32) -- result of executing the ADD operation

-- What does it mean to execute the ADD opcode?
def execute_RTYPE_add_pure (input : AddInput) : AddOutput := {
  -- next program counter is always the current one + 4
  nextPC := input.PC + 4#32
  -- the result of executing the operation is either:
  rd := if h : input.rd = 0
    -- no result, if the intended output register equals zero
    then .none
    -- bit-vector addition, which is stored in register rd, which we now know is non-zero
    else .some (
      ⟨ input.rd.val, by apply Finset.mem_Icc_mpr; omega ⟩,
      input.r1_val + input.r2_val
    ) : AddOutput
}

-- Equivalence between actual RISC-V execution of ADD and execute_RTYPE_add_pure
lemma execute_RTYPE_add_pure_equiv
  -- For a given AddInput
  (add_input : AddInput)
  -- and registers r1, r2, and rd, such that
  (r1 r2 rd : regidx)
  -- the first operand of the AddInput is stored in register r1 in the RISC-V state,
  (h_input_r1 : read_xreg (regidx_to_fin r1) state = EStateM.Result.ok (add_input.r1_val) state)
  -- the second operand of the AddInput is stored in register r2 in the RISC-V state,
  (h_input_r2 : read_xreg (regidx_to_fin r2) state = EStateM.Result.ok (add_input.r2_val) state)
  -- the intended output register of the AddInput is rd,
  (h_input_rd : add_input.rd = regidx_to_fin rd)
  -- and the program counter of the AddInput is stored in the PC register in the RISC-V state
  (h_input_pc : state.regs.get? Register.PC = .some add_input.PC)
: -- then
  -- executing the RISC-V add on a given state
  ( do
```

```

Sail.writeReg Register.nextPC (Sail.BitVec.addInt (← Sail.readReg Register.PC) 4)
LeanRV32D.Functions.execute (instruction.RTYPE (r2, r1, rd, rop.ADD))
) state
-- is equivalent to, on the same state
=
-- running execute_RTYPE_add_pure
let add_output : AddOutput := execute_RTYPE_add_pure add_input
( do
  -- storing the nextPC of the AddOutput in the PC register
  Sail.writeReg Register.nextPC add_output.nextPC
  -- and updating the intended output register with the sum of the two operands
  -- if the intended output register does not equal zero
  match add_output.rd with
  | .some (rd, rd_val) => write_xreg rd rd_val
  | .none => pure ()
  pure (ExecutionResult.Retire_Success ())
) state

```

This process for the ADDI opcode is similar (cf. [addi.lean](#)), and below we highlight the differences only:

```

structure AddInput where
  r1_val : BitVec 32
  imm    : BitVec 12 -- 12-bit immediate value
  rd     : Fin 32
  PC     : BitVec 32

structure AddOutput where
  nextPC : BitVec 32
  rd      : Option (Finset.Icc 1 31 × BitVec 32)

def execute_ITYPE_addi_pure (input : AddInput) : AddOutput := {
  nextPC := input.PC + 4#32
  rd := if h: input.rd = 0
    then .none
    else .some (
      ⟨ input.rd.val, by apply Finset.mem_Icc.mpr; omega ⟩,
      -- bit-vector addition of the first operand and the sign-extended 12-bit immediate
      input.r1_val + BitVec.signExtend 32 input.imm
    ) : AddOutput
}

lemma execute_ITYPE_addi_pure_equiv
  (add_input : AddInput)
  -- 12-bit immediate value
  (imm : BitVec 12)
  (r1 rd : regidx)
  (h_input_r1: read_xreg (regidx_to_fin r1) state = EStateM.Result.ok (add_input.r1_val) state)
  -- must match the AddInput immediate
  (h_input_imm: add_input.imm = imm)
  (h_input_rd: add_input.rd = regidx_to_fin rd)
  (h_input_pc: state.regs.get? Register.PC = .some add_input.PC)
:
( do
  Sail.writeReg Register.nextPC (Sail.BitVec.addInt (← Sail.readReg Register.PC) 4)
  LeanRV32D.Functions.execute (instruction.ITYPE (imm, r1, rd, iop.ADDI))
) state =
let addi_output := execute_ITYPE_addi_pure add_input
( do
  Sail.writeReg Register.nextPC addi_output.nextPC
  match addi_output.rd with
  | .some (rd, rd_val) => write_xreg rd rd_val
  | .none => pure ()
  pure (ExecutionResult.Retire_Success ())
) state

```

Row-level equivalence. Next, we move to the [BaseALU.lean](#) equivalence file, in which we first define a structure that contains all of the information required to reason about a row of the base ALU chip, including the bus contents:

```
structure ALU_instruction_fields where
  is_valid : FBB
  non_imm : FBB

  pc : FBB
  next_pc : FBB

  opcode : FBB

  prev_a_timestamp : FBB
  a_timestamp : FBB
  prev_b_timestamp : FBB
  b_timestamp : FBB
  prev_c_timestamp : FBB
  c_timestamp : FBB
  timestamp : FBB
  next_timestamp : FBB

  rs1_ptr : FBB
  rs2_ptr : FBB
  rd_ptr : FBB

  prev_a : Vector FBB 4
  a : Vector FBB 4
  b : Vector FBB 4
  c : Vector FBB 4

  range_checked_vals : Vector FBB 6
  bitwise_vals : Vector (Vector FBB 3) 5
```

We next define how to extract the opcode inputs from `ALU_instruction_fields`, given below for `ADD` and `ADDI`:

```
def AddInput_of_ALU_instruction_fields (row : ALU_instruction_fields) : PureSpec.AddInput := {
  r1_val := BabyBear.toBV32 row.b
  r2_val := BabyBear.toBV32 row.c
  rd := wrap_to_regidx row.rd_ptr
  PC := row.pc.toNat
  : PureSpec.AddInput
}

def AddiInput_of_ALU_instruction_fields (row : ALU_instruction_fields) : PureSpec.AddiInput := {
  r1_val := BabyBear.toBV32 row.b
  imm := BitVec.ofNat 12 row.rs2_ptr.toNat
  rd := wrap_to_regidx row.rd_ptr
  PC := row.pc.toNat
  : PureSpec.AddiInput
}
```

where

```
def wrap_to_regidx (val : FBB) : Fin 32 :=
  ⟨val / 4 % 32, by grind⟩
```

and also what it means for the opcode output to match given `ALU_instruction_fields`:

```
def AddOutput_matches_ALU_instruction_fields
  (row : ALU_instruction_fields) (add_output : PureSpec.AddOutput)
: Prop :=
  BabyBear.isU32 row.b ∧ -- b must represent a 32-bit integer
  BabyBear.isU32 row.c ∧ -- c must represent a 32-bit integer
  add_output.nextPC = row.next_pc.toNat ∧ -- the next PCs must match
  match add_output.rd with
```

```

-- if the output is none, the output register must be 0
| .none => row.a = row.prev_a ^ row.rd_ptr = 0
-- otherwise
| .some (rd, rd_val) =>
  -- a must represent a 32-bit integer
  BabyBear.isU32 row.a ^
  -- it must be equal to the result of the addition
  BabyBear.toBV32 row.a = rd_val ^
  -- and the output registers must match
  rd.1.toNat * 4 = row.rd_ptr.toNat

def AddiOutput_matches_ALU_instruction_fields
  (row : ALU_instruction_fields) (addi_output : PureSpec.AddiOutput)
: Prop :=
  -- b must represent a 32-bit integer
  BabyBear.isU32 row.b ^
  -- the immediate value must be sign-extended correctly
  (BitVec.ofNat 12 (row.rs2_ptr)).signExtend 24 = (BitVec.ofNat 24 (row.rs2_ptr)) ^
  -- the next PCs must match
  addi_output.nextPC = row.next_pc.toNat ^
  -- the same as for ADD
  match addi_output.rd with
  | .none => row.a = row.prev_a ^ row.rd_ptr = 0
  | .some (rd, rd_val) =>
    BabyBear.isU32 row.a ^
    BabyBear.toBV32 row.a = rd_val ^
    rd.1.toNat * 4 = row.rd_ptr.toNat

```

Next, we give a correctness specification for all of the base ALU opcodes in terms of opcode inputs, opcode outputs, and `ALU_instruction_fields`:

```

def ALU_instruction_fields.spec (row : ALU_instruction_fields) : Prop :=
  row.is_valid = 1 → (
    -- for valid rows
    (row.non_imm = 0 ∨ row.non_imm = 1) ^ -- non_imm is a flag
    row.opcode ∈ Finset.Icc 512 516 ^ -- the opcodes are between 512 and 516 inclusive
    -- non-immediate opcode variants
    (row.non_imm = 1 → (
      -- correctness of ADD
      (row.opcode = 512 →
        AddOutput_matches_ALU_instruction_fields row
        (PureSpec.execute_RTYPE_add_pure (AddInput_of_ALU_instruction_fields row)))
      ) ^
      ...
    )) ^
    -- immediate opcode variants
    (row.non_imm = 0 → (
      -- correctness of ADDI
      (row.opcode = 512 →
        AddiOutput_matches_ALU_instruction_fields row
        (PureSpec.execute_ITYPE_addi_pure (AddiInput_of_ALU_instruction_fields row)))
      ) ^
      ...
    ))
  )

```

Next, we express the content put on the buses by the base ALU chip in terms of `ALU_instruction_fields` (e.g.:

```

def ALU_instruction_fields.execution (row : ALU_instruction_fields) : List (FBB × List FBB) := [
  (-row.is_valid, [row.pc, row.timestamp]),
  (row.is_valid, [row.next_pc, row.next_timestamp])
]

```

for the execution bus—for others, see [BaseALU.lean](#)) and define what it means that we have done this correctly:

```

def bus_from_instruction_fields (rows : List ALU_instruction_fields) : ℕ → List (FBB × List FBB) :=

```



```

λ index ⇒
  if index = ExecutionBus then      rows.flatMap ALU_instruction_fields.execution
  else if index = MemoryBus then    rows.flatMap ALU_instruction_fields.memory
  else if index = RangeCheckerBus then rows.flatMap ALU_instruction_fields.range_checks
  else if index = ProgramBus then   rows.flatMap ALU_instruction_fields.read_instruction
  else if index = BitwiseBus then   rows.flatMap ALU_instruction_fields.bitwise
  else []

```

Next, we define how to construct `ALU_instruction_fields` given a row of the base ALU trace:

```

def get_instruction_fields_row (air : Valid_VmAirWrapper_alu FBB ExtF) (row : ℕ) : ALU_instruction_fields := {
  is_valid := air.core.is_valid row 0
  non_imm := air.adapter.rs2_as row 0
  pc := air.adapter.from_state.pc row 0
  next_pc := air.adapter.from_state.pc row 0 + 4
  opcode := (air.core.ctx row 0).instruction.opcode
  prev_a_timestamp := air.adapter.writes_aux.base.prev_timestamp row 0
  a_timestamp := air.adapter.from_state.timestamp row 0 + 2
  prev_b_timestamp := air.adapter.reads_aux_0.base.prev_timestamp row 0
  b_timestamp := air.adapter.from_state.timestamp row 0
  prev_c_timestamp := air.adapter.reads_aux_1.base.prev_timestamp row 0
  c_timestamp := air.adapter.from_state.timestamp row 0 + 1
  timestamp := air.adapter.from_state.timestamp row 0
  next_timestamp := air.adapter.from_state.timestamp row 0 + 3
  rs1_ptr := air.adapter.rs1_ptr row 0
  rs2_ptr := air.adapter.rs2_ptr row 0
  rd_ptr := air.adapter.rd_ptr row 0
  prev_a := #v[air.adapter.writes_aux.prev_data_0 row 0, air.adapter.writes_aux.prev_data_1 row 0,
    air.adapter.writes_aux.prev_data_2 row 0, air.adapter.writes_aux.prev_data_3 row 0]
  a := #v[air.core.a_0 row 0, air.core.a_1 row 0, air.core.a_2 row 0, air.core.a_3 row 0]
  b := #v[air.core.b_0 row 0, air.core.b_1 row 0, air.core.b_2 row 0, air.core.b_3 row 0]
  c := #v[air.core.c_0 row 0, air.core.c_1 row 0, air.core.c_2 row 0, air.core.c_3 row 0]
  range_checked_vals :=
    #v[air.adapter.reads_aux_0.base.timestamp_lt_aux.lower_decomp_0 row 0,
      air.adapter.reads_aux_0.base.timestamp_lt_aux.lower_decomp_1 row 0,
      air.adapter.reads_aux_1.base.timestamp_lt_aux.lower_decomp_0 row 0,
      air.adapter.reads_aux_1.base.timestamp_lt_aux.lower_decomp_1 row 0,
      air.adapter.writes_aux.base.timestamp_lt_aux.lower_decomp_0 row 0,
      air.adapter.writes_aux.base.timestamp_lt_aux.lower_decomp_1 row 0]
  bitwise_vals :=
    #v[ #v[air.core.x_0 row 0, air.core.y_0 row 0, air.core.x_xor_y_0 row 0],
      #v[air.core.x_1 row 0, air.core.y_1 row 0, air.core.x_xor_y_1 row 0],
      #v[air.core.x_2 row 0, air.core.y_2 row 0, air.core.x_xor_y_2 row 0],
      #v[air.core.x_3 row 0, air.core.y_3 row 0, air.core.x_xor_y_3 row 0],
      #v[air.core.c_0 row 0, air.core.c_1 row 0, 0] ]
  : ALU_instruction_fields
}

```

What did we actually prove? Our main theorem for the base ALU chip is as follows:

```

theorem base_ALU_spec [Field ExtF]
  (air : Valid_VmAirWrapper_alu FBB ExtF)
  (h_constraints : allHold_allRows air)
  (h_bus_ax : ∀ row ≤ air.last_row, VmAirWrapper_alu.constraints.axiomsPerRow air row)
  (h_bus_wf : ∀ row ≤ air.last_row, VmAirWrapper_alu.constraints.wf_propertiesToAssumePerRow air row)
  :
  ∃ instruction_fields_list : List ALU_instruction_fields,
    air.buses = bus_from_instruction_fields instruction_fields_list ∧
    instruction_fields_list.Forall ALU_instruction_fields.spec
:= by
  use get_instruction_fields air
  simp [
    bus_from_instruction_fields_eq_air_buses air h_constraints,
    spec_of_get_instruction_fields air h_constraints h_bus_ax h_bus_wf
  ]

```

]

which tells us that, given a base ALU air wrapper `air`, if, on all rows of the trace:

- all of the constraints hold (`h_constraints`);
- all of the bus axioms hold (`h_bus_ax`); and
- all of the bus assumptions hold (`h_bus_ax`)

then we can find a list of `ALU_instruction_fields` such that, for each row of the trace:

- it matches the entries put on the buses by that row; and
- if the row is valid, the behaviour of the zkVM is equivalent to the behaviour of the official Lean RISC-V spec, as illustrated, e.g., by the `execute_RTYPE_add_pure_equiv` and `execute_RTYPE_addi_pure_equiv` lemmas.

In the proof, as the witness we choose

```
def get_instruction_fields (air : Valid_VmAirWrapper_alu FBB ExtF) : List ALU_instruction_fields :=
  (List.range (air.last_row + 1)).map (λ row => get_instruction_fields_row air row)
```

that is, a mapping of the above-presented `get_instruction_fields_row` function across all rows. This, together with the definition of `get_instruction_fields_row`, gives us the equivalence guarantees for the entire base ALU chip.

3.2 Example 2: Control-flow opcodes

The verification process of control-flow opcodes is, for the most part, entirely analogous to that of ALU-related opcodes. The main point of divergence is that control-flow opcodes may affect the program counter in more complex ways and interact more with the RISC-V state. We illustrate this on the BEQ example, jumping directly into the last of the verification (as the previous steps do not have notable differences), in which we link the proven pure specification to the structural part of the Lean RISC-V opcode specification.

The pure specification of BEQ captures the expected branching, and is as follows:

```
include
  row_valid constraints axioms propertiesToAssume
in
theorem spec_BEQ_BNE_pc
:
  -- BEQ case
  (air.core.expected_opcode row 0 = 544 →
    BitVec.ofNat 32 (air.to_pc row 0) =
      if U32.toBV #v[(air.core.a_0 row 0).val, (air.core.a_1 row 0).val,
        (air.core.a_2 row 0).val, (air.core.a_3 row 0).val] ==
        U32.toBV #v[(air.core.b_0 row 0).val, (air.core.b_1 row 0).val,
        (air.core.b_2 row 0).val, (air.core.b_3 row 0).val]
      -- operands equal, jump to pc + immediate
      then BitVec.ofNat 32 (air.adapter.from_state.pc row 0).val +
        BitVec.signExtend 32 (BitVec.ofInt 13 (BabyBear.toInt (air.core.imm row 0)))
      -- operands not equal, proceed to next instruction
      else BitVec.ofNat 32 (air.adapter.from_state.pc row 0).val + 4#32) ^
  ...
```

The structures that capture the input and the output for BEQ are defined as follows:

```
structure BeqInput where
  imm.   : BitVec 13 -- 13-bit immediate value
  r1_val : BitVec 32 -- first operand of BEQ
  r2_val : BitVec 32 -- second operand of BEQ
  PC     : BitVec 32 -- current program counter
  misa   : BitVec 32 -- misa register of the RISC-V machine

structure BeqOutput where
  nextPC : BitVec 32 -- next program counter
  success : Bool     -- success indicator
```

```

throws : Bool      -- exception indicator

def execute_BEQ_pure (input : BeqInput) : BeqOutput :=
  -- non-success checks do not need to be performed if operands are not equal
  let skip := !(input.r1_val == input.r2_val)
  -- BEQ throws if the operands are equal and next PC is not divisible by 2
  let throws := !skip && BitVec.ofBool (input.PC + BitVec.signExtend 32 input.imm)[0] == 1#1
  -- BEQ fails either if it throws or if the operands are equal and the next PC is not divisible by 4
  -- but the misa register indicates that compression (2-byte aligned PCs) is not allowed
  let fails := throws ||
    (!skip &&
      BitVec.ofBool (input.PC + BitVec.signExtend 32 input.imm)[1] == 1#1 && !input.misa[2])
  {
    -- The program counter moves to either
    nextPC := if skip || fails
      -- the next instruction, if operands are not equal or BEQ fails
      then (input.PC + 4)
      -- or to the sum of the program counter and the immediate value
      else (input.PC + BitVec.signExtend 32 input.imm)
    success := !fails -- BEQ succeeds if it does not fail
    throws := throws
    : BeqOutput
  }

```

and we, analogously to the ADD/ADDI, prove the equivalence between the actual RISC-V execution of BEQ and `execute_BEQ_pure`, as follows:

```

lemma execute_BEQ_pure_equiv
  (beq_input : BeqInput)
  (imm: BitVec 13)
  (h_input_imm: beq_input.imm = imm)
  (h_input_r1: read_xreg (regidx_to_fin r1) state = EStateM.Result.ok (beq_input.r1_val) state)
  (h_input_r2: read_xreg (regidx_to_fin r2) state = EStateM.Result.ok (beq_input.r2_val) state)
  (h_input_pc: state.regs.get? Register.PC = .some beq_input.PC)
  (h_input_misa: state.regs.get? Register.misa = .some beq_input.misa)
  :
  ( do
    Sail.writeReg Register.nextPC (Sail.BitVec.addInt (← Sail.readReg Register.PC) 4)
    LeanRV32D.Functions.execute (instruction.BTYPE (imm, r2, r1, bop.BEQ))
  ) state =
  let beq_output := execute_BEQ_pure beq_input
  ( do
    -- update the program counter with the next computed value in all cases
    Sail.writeReg Register.nextPC beq_output.nextPC
    if beq_output.throws then
      -- exceptional case: next PC not divisible by two
      throw (Sail.Error.Assertion "extensions/I/base_insts.sail:59.29-59.30")
    else if !beq_output.success then
      -- failing case: next PC divisible by two, but not divisible by four
      pure (
        ExecutionResult.Memory_Exception (
          (virtaddr.Virtaddr (beq_input.PC + BitVec.signExtend 32 beq_input.imm)),
          (ExceptionType.E_Fetch_Addr_Align ())
        )
      )
    else
      -- success case
      (pure (ExecutionResult.Retire_Success ()))) state

```

noting how we break down and make explicit the control flow into the success, failure, and exception cases. Importantly, as the transpilation process ensures that all program counters used in the execution are divisible by four (cf. 2.3.1), the throw and failure cases do not get triggered in OpenVM. The remaining proofs are analogous.

3.3 Example 3: LOADW and STOREW

The Lean RISC-V specification is particularly complex when it comes to the memory manipulation. As a result of this, and unlike for other opcodes, for memory-manipulating opcodes we need to make additional assumptions on the RISC-V state and provide four axioms as a result of the current state of the Lean RISC-V specification. The assumptions are effectively the same ones as made by Succinct Labs in their formal verification of the 64-bit SP1 HyperCube.² These assumptions and axioms are contained to the part of the verification in which we connect to the structural part of the Lean RISC-V opcode specification. As they are the same for LOADW and STOREW, and so we illustrate these assumptions only using LOADW.

As for the previous examples, we start by describing the input and the output of LOADW:

```
structure LwInput where
  -- operands
  r1 : BitVec 5 -- register of first operand
  imm : BitVec 12 -- 12-bit immediate
  rd : BitVec 5 -- intended output register
  -- registers
  r1_val : BitVec 32 -- value stored in register of first operand
  PC : BitVec 32 -- current program counter
  -- four bytes, from lowest to highest, that are loaded from the memory
  data0 : BitVec 8
  data1 : BitVec 8
  data2 : BitVec 8
  data3 : BitVec 8
  -- RISC-V registers on which assumptions are required
  mstatus : BitVec 64
  cur_privilege : Privilege
  plat_clint_base : BitVec 34
  plat_clint_size : BitVec 34
  plat_ram_base : BitVec 34
  plat_ram_size : BitVec 34
  plat_rom_base : BitVec 34
  plat_rom_size : BitVec 34
  htif_tohost_base : Option (BitVec 34)

structure LwOutput where
  nextPC : BitVec 32 -- next program counter
  rd : Option (Finset.Icc 1 31 × BitVec 32) -- result of executing the LOADW opcode
```

and give the assumptions required for proving equivalence, providing explanations for some of them in comments:

```
def lw_state_assumptions
  (i : LwInput)
  (state : PreSail.SequentialState RegisterType Sail.trivialChoiceSource)
: Prop :=
  -- Standard assumptions, simply connecting the RISC-V registers to the
  -- fields of the input structure, as for ADD/ADDI/BEQ/other opcodes
  LeanRV32D.Functions.rX_bits (regidx.Regidx i.r1) state = EStateM.Result.ok i.r1_val state ∧
  Sail.readReg Register.mstatus state = EStateM.Result.ok i.mstatus state ∧
  Sail.readReg Register.cur_privilege state = EStateM.Result.ok i.cur_privilege state ∧
  Sail.readReg Register.plat_clint_base state = EStateM.Result.ok i.plat_clint_base state ∧
  Sail.readReg Register.plat_clint_size state = EStateM.Result.ok i.plat_clint_size state ∧
  Sail.readReg Register.plat_ram_base state = EStateM.Result.ok i.plat_ram_base state ∧
  Sail.readReg Register.plat_rom_base state = EStateM.Result.ok i.plat_rom_base state ∧
  Sail.readReg Register.plat_ram_size state = EStateM.Result.ok i.plat_ram_size state ∧
  Sail.readReg Register.plat_rom_size state = EStateM.Result.ok i.plat_rom_size state ∧
  Sail.readReg Register.htif_tohost_base state = EStateM.Result.ok i.htif_tohost_base state ∧
  state.mem[(i.r1_val + BitVec.signExtend 32 i.imm).toNat]? = .some i.data0 ∧
  state.mem[(i.r1_val + BitVec.signExtend 32 i.imm).toNat + 1]? = .some i.data1 ∧
  state.mem[(i.r1_val + BitVec.signExtend 32 i.imm).toNat + 2]? = .some i.data2 ∧
  state.mem[(i.r1_val + BitVec.signExtend 32 i.imm).toNat + 3]? = .some i.data3 ∧
  state.regs.get? Register.PC = .some i.PC ∧
```

²This formal verification effort was, in principle, joint between Nethermind and Succinct Labs, but the verification of memory-related opcodes was done exclusively by Succinct Labs.

```
-- Additional assumptions (AS1-AS9) on the RISC-V state
LeanRV32D.Functions.is_aligned_vaddr (virtaddr.VirtAddr (i.r1_val + BitVec.signExtend 32 i.imm)) 4 = true ∧
(i.r1_val + BitVec.signExtend 32 i.imm).toNat + 4 < i.plat_ram_size ∧
i.cur_privilege = Privilege.Machine ∧
BitVec.extractLsb 17 17 i.mstatus = 0#1 ∧
i.plat_clint_base = 0 ∧
i.plat_clint_size = 0 ∧
i.plat_ram_base = 0 ∧
i.plat_rom_base = 0 ∧
i.htif_tohost_base = .none
```

We now discuss the purpose of each of the additional assumptions and axioms (AX1-AX4). All four axioms are sorried out due to the deficiencies in the Lean RISC-V specification itself, where definitions are given using the for and repeat constructs, which cannot be reasoned about due to their partiality. We are in communication with the specification developers, understanding how this can be addressed.

AS1: Alignment. This assumption states that the address from which the load is to be performed should be aligned to a multiple of four. This was requested by OpenVM developers, as the correctness guarantees that they aim to provide are modulo this assumption, which should be maintained by the user of the zkVM.

AS2: Overflow. This assumption states that the load does not overflow the available memory.

AS3: Privilege. This assumption states that the privilege with which the RISC-V code should be executed is machine-mode. This, in a sense, corresponds to a bare-metal execution within which one is able to modify any part of the RISC-V state.

AS4: Machine-mode load and store. This assumptions states that the MPRV bit of the mstatus RISC-V register is set to zero. This means that memory loads and stores are handled with machine-mode privilege, rather than the privilege level given in the MPP (bits 11-12 of mstatus).

AS5-AS6: CLINT. These assumptions bypass the CLINT (Core-Local Interrupt Controller), which is not part of the core ISA specification and is not modelled by OpenVM, but is modelled by the Lean RISC-V specification.

AS7-AS8: ROM and RAM starting addresses. These assumptions state that both the RISC-V ROM and the RAM start from address 0.

AS9: Host-target interface. This assumption states that there the host-target interface is not used.

AX1-AX2: Simplified implementations of vmem_read and vmem_write. This axiom states that our reformulation of the vmem_read and vmem_write functions are equivalent to the original implementations:

```
-- 'axiom' until the Lean RISC-V spec is rebuilt to avoid the repeat
lemma vmem_read'_equiv :
  LeanRV32D.Functions.vmem_read = vmem_read'
:= by
  sorry

-- 'axiom' until the Lean RISC-V spec is rebuilt to avoid repeat
lemma vmem_write_addr'_equiv :
  LeanRV32D.Functions.vmem_write_addr = vmem_write_addr'
:= by
  sorry
```

AX3-AX4: Handling of PMP checks. These two axioms state that the PMP checks, which are required for every memory management action, pass.

```
/-- We can't prove this directly because the loop in 'pmpCheck' doesn't unfold.
    Adding this is at least consistent, since the left-hand side has no actual value. -/
lemma pmp_check_machine_read (reg_val : BitVec 32) (offset : BitVec 32) (width : ℕ) (s) :
```

```

EStateM.run (
  LeanRV32D.Functions.pmpCheck (
    physaddr.Physaddr (
      LeanRV32D.Functions.zero_extend (
        Sail.BitVec.addInt (reg_val + offset) 0
      )
    )
  )
  width
  (AccessType.Read ())
  Privilege.Machine
) s = EStateM.Result.ok none s
:= by
  sorry

/-- We can't prove this directly because the loop in 'pmpCheck' doesn't unfold.
    Adding this is at least consistent, since the left-hand side has no actual value. -/
lemma pmp_check_machine_write (reg_val : BitVec 32) (offset : BitVec 32) (width : N) (s) :
  EStateM.run (
    LeanRV32D.Functions.pmpCheck (
      physaddr.Physaddr (
        LeanRV32D.Functions.zero_extend (
          Sail.BitVec.addInt (reg_val + offset) 0
        )
      )
    )
    width
    (AccessType.Write ())
    Privilege.Machine
  ) s = EStateM.Result.ok none s
:= by
  sorry

```

4 Assumptions and Caveats

The outcomes of this engagement are subject to the following assumptions and caveats:

1. the correctness of Lean's kernel;
2. the correctness of the RISC-V Lean specification with respect to the official RISC-V standard;
3. the correctness of the theory and implementation of the proof system utilised by OpenVM;
4. the correctness of the theory and implementation of the OpenVM lookup argument and bus interaction mechanism, including memory consistency;
5. for the correctness proofs of LOADW and STOREW, as per §3.3:
 - (a) that the address provided for LOADW and STOREW is aligned to a multiple of four;
 - (b) that the loading/storing does not access addresses beyond the memory bound (2^{29} for OpenVM);
 - (c) that the RISC-V execution is run with machine-mode privilege, and that the MPRV bit is not set;
 - (d) that the RISC-V CLINT mechanism is not used;
 - (e) that the RISC-V ROM and RAM address spaces start from zero;
 - (f) that there is no RISC-V host-target interface used;
 - (g) that the four axiomatised lemmas on RISC-V auxiliary functions hold, which will be provable once the official RISC-V specification has been updated to use looping constructs that can be reasoned about.

5 Disclaimer

This Report is based on the scope of materials and documentation provided by you to Nethermind in order that Nethermind could conduct the tasks outlined in the Introduction. The content of this Report is intended to provide an analysis of the formal verification process undertaken, and the results obtained, based on the information and data available to Nethermind up to the time of completion. The formal verification process conducted by the team of engineers was based on the specific parameters, methodologies, and data available at the time of the analysis. Any changes in the underlying parameters, methodologies, or data may lead to different results.

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